Accident Analysis and Prevention 215 (2025) 108011

Contents lists available at ScienceDirect

Accident Analysis and Prevention

journal homepage: www.elsevier.com/locate/aap

Safety validation for connected autonomous vehicles using large-scale testing tracks in high-fidelity simulation environment☆

Zheng Xu [a,*], Xiaomeng Wang [b], Xuesong Wang [c], Nan Zheng [a]

[a] Department of Civil Engineering, Monash University, Melbourne, VIC 3800, Australia
[b] College of Transport and Communications, Shanghai Maritime University, Shanghai, China
[c] School of Transportation Engineering, Tongji University, Shanghai 201804, China

ARTICLE INFO

Keywords:
Autonomous driving system
Connected autonomous vehicles
Safety validation
Large-scale testing tracks
High-fidelity simulation

ABSTRACT

Public concern over the implementation of Connect Autonomous Vehicles (CAVs) remains a significant issue, and safety validation for CAVs remains a critical challenge due to the limitations of existing testing methods. While real-world testing is crucial, it can be expensive, time-consuming, and potentially impractical for evaluating the operation of CAV fleets. This paper presents a comprehensive co-simulation framework integrating the fully compiled CARLA with traffic microsimulation to establish a large-scale (20×20 km^2) testing environment for systematic CAV safety validation. The framework encompasses three key components: 1) a high-fidelity testing environment featuring diverse road geometries and dynamic conditions including weather variations and realistic traffic flows; 2) an intelligent CAV function developed through deep reinforcement learning and enhanced with utility-based connectivity strategies; 3) A sophisticated safety measurement metric that utilizes surrogate safety assessments, integrating a multi-type Bayesian hierarchical model to comprehensively evaluate risk factors and incident probabilities. The case study assessed CAV penetration rates ranging from 0 % to 100 %, identifying an optimal safety performance at a 70 % penetration rate, which resulted in an 86.05 % reduction in accident rates compared to conventional driving scenarios. This optimal safety level was effectively achieved in rural and suburban areas, where the average conflict probability was 0.4. However, in transition zones that connect high-, medium-, and low-density areas, significant traffic conflicts persisted even at this optimal penetration rate, with a conflict probability exceeding 0.7. Key results highlight critical safety patterns under optimal conditions, revealing that roundabouts and signalized intersections account for over 70 % of conflicts involving CAVs. This work advances CAV safety validation by providing a more realistic, large-scale testing environment that compensates for real-world testing limitations and allows for comprehensive safety evaluations across diverse scenarios.

1. Introduction

The commercial deployment of Connected and Autonomous Vehicles (CAVs) has witnessed unprecedented acceleration worldwide, marking a transformative shift in transportation systems. Major automotive manufacturers and technology companies have made significant strides in bringing Level 4 autonomous vehicles (AVs) (SAE Interntionl, 2018) to market (Rana and Hossain, 2023). In the US, Europe, and Asia, substantial regulatory progress has facilitated the expanded testing and deployment of AVs and CAVs on public roads (Hansson, 2020). Although this advancement is evident in various metropolitan areas, it has also highlighted critical concerns regarding safety validation methodologies. This is particularly pressing as current testing approaches largely depend on limited-scale real-world pilots or simplified simulation environments (Koopman and Wagner, 2016; Feng et al., 2021; Rahman and Thill, 2023). Furthermore, this rapid expansion has not been without challenges, as evidenced by reported incidents involving AVs. Various studies have documented accidents, ranging from minor conflicts to serious collisions, particularly at intersections with conventional vehicles (CV) and vulnerable road users (Xu et al., 2019; Alozi and Hussein, 2022; Fu et al., 2021; Malik et al., 2020). These incidents highlight the complexities involved in validating the safety of

☆ This article is part of a special issue entitled: ‘Safety validation’ published in Accident Analysis and Prevention.

* Corresponding author.
E-mail address: Zheng.xu3@monash.edu (Z. Xu).

https://doi.org/10.1016/j.aap.2025.108011
Received 31 October 2024; Received in revised form 11 February 2025; Accepted 12 March 2025
Available online 18 March 2025

autonomous driving system (ADS) across diverse scenarios, weather conditions, and traffic interactions. This calls for more comprehensive testing frameworks that can effectively bridge the gap between controlled testing environments and real-world deployment while ensuring public safety (Yan et al., 2023; Yang et al., 2024).

In literature, studies focusing on ADS testing and safety validation have evolved along three primary approaches: simulation-based approaches, small-scale testing tracks, and controlled public road testing (Tang et al., 2023). Simulation-based methods are particularly popular due to their cost-effectiveness and capability to replicate diverse scenarios. For instance, platforms like CARLA offer sophisticated testing environments, which increasingly include high-fidelity rendering to improve fidelity and flexibility (Dosovitskiy et al., 2017). An example of this application is the work of Li et al. (2022), who assessed a risk-aware function on a 420 m road segment using the CARLA platform, with vehicles approaching the ADS to evaluate performance metrics such as the driving distance before collisions. Similarly, Abdel-Aty et al. (2022) implemented an Automated Emergency Braking system in CARLA, integrating a sensor model for pedestrian detection and a braking strategy based on safety indicator thresholds, enhanced by a LiDAR sensor with configurable Field of View parameters, across 10,368 simulated cases.

Genuine driving behavior data in real traffic conditions without experimental control is defined as naturalistic driving data (Simmons et al., 2016; Caird et al., 2018; Wijayaratna et al., 2019). Obtaining naturalistic driving data is considered the most direct method to physically validate vehicle safety performance. Numerous studies are pursuing this approach. However, safety testing for CAV fleets in naturalistic driving scenarios not only poses risks to public roads, involving real people and property, but also challenges existing traffic control measures and the safety of other road users. Small-scale testing tracks provide physical validation but are often confined to areas less than 5 km^2, limiting the variety of testable scenarios and environmental conditions. For instance, Kolekar et al. (2021) examined user adaptation to ADS on a controlled testing track, revealing spontaneous feedback on risk scenarios at a moderate driving speed of 40 km/h in a naturalistic setting. Xu et al. (2021) conducted ADS testing on a track combining a 2.4 km two-lane circular track and a 1.1 km straight four-lane track, engaging 261 participants in eight specific scenarios to gauge user feedback on potential ADS failures.

Controlled public road testing provides real-world validation but comes with significant challenges related to safety risks, regulatory compliance, and limitations in systematically testing edge cases. For example, Yu and Li (2022) analyzed operational test data from commercialized AVs in Shanghai, focusing on vehicle kinematic data such as speed, acceleration, and jerk to understand the relationship between driving environment variables and vehicle behaviors in hazardous scenarios. Liu et al. (2024) reviewed 197 ADS-involved crash reports from public roads in California spanning 2014 to 2022, examining pre-crash movements and decision-making processes, particularly regarding stop decisions, deceleration, and collision avoidance, to identify the patterns of potential flaws in ADS algorithms that may have contributed to crashes.

Recent efforts to bridge various testing methodologies have led to the development of co-simulation strategies that integrate traffic microsimulation with high-fidelity virtual environments (Chen et al., 2024; Xu et al., 2024a) or with naturalistic driving environments (Feng et al., 2020). Specifically, Chen et al. (2024) created a software-in-the-loop co-simulation platform that combines PreScan for environmental perception and AV sensor modelling, Matlab for decision-making and control models, and traffic microsimulation for a realistic traffic flow environment. This platform was designed to evaluate AV safety performance in freeway merging areas, overcoming the limitations of single simulation software by merging multiple simulation strategies for a comprehensive testing approach. Xu et al. (2024a) developed a co-simulation strategy that merges traffic microsimulation with CARLA, focusing on generating critical scenarios involving ADS in high-fidelity environments. This approach enhances data synthesis and quantifies the discrepancy in risk perception between humans and ADS. Feng et al. (2020) introduced an augmented reality (AR) testing framework that combines a physical test track with a microsimulation environment. In this setup, real autonomous vehicles could operate on a physical track and interact with virtual background vehicles generated in the simulation. Yan et al. (2023) proposed an integrated structure that includes a transformer-based behavior modelling network for multi-agent interactions, a conflict critic module to manage safety–critical events, and a safety mapping network that corrects dangerous behaviors. This framework more accurately reproduces both normal driving and crash statistics and is scalable to larger traffic networks, enhancing the validity of ADS testing frameworks.

Many existing validation methodologies focus solely on the testing of individual ADS, struggling to comprehensively assess CAV performance across a wide range of operational scenarios. This includes complex interactions with various road users and diverse environmental conditions (Ni et al., 2024; Sohrabi et al., 2021). Few studies have specifically addressed the safety validation of CAV operations. For example, Jin et al. (2018) conducted experiments on both closed tracks and public roads. In their tests, a CAV received motion information from 2 to 3 human-driven vehicles ahead via V2X communication. On the closed track, they evaluated safety–critical scenarios such as beyond-line-of-sight braking and cascade braking events. On public roads, they tested over an 8-mile segment to assess how well the CAV could mitigate traffic waves affecting a string of 6 human-driven vehicles ahead. Performance metrics included minimum speeds, acceleration distributions, and energy consumption, comparing various control strategies based on the number of vehicles ahead (1–3 vehicles). Additionally, Pereira et al. (2019) implemented a testing framework on a motorway in Portugal, where CAVs operated between two traffic control vehicles within a “moving corridor” and received safety notifications through roadside units (RSUs). This test evaluated the CAVs’ responses to different safety events broadcast via decentralized environmental notification messages, particularly focusing on how vehicles adjusted their speed in response to these notifications.

As pointed out by Lee and Hess (2020), safety validation for CAV fleets is more demanding than for single ADS. The feasibility of systematically testing CAV performance through real-world observational studies is relatively low. The opportunities for legally permitted CAV testing are limited, requiring appropriate regulatory frameworks that are often not in place for real-world scenarios. Furthermore, as Khan et al. (2023) highlighted, simulation strategies are typically case-specific and offer limited interaction fidelity. This situation emphasizes the need for innovative methods that can address the challenges of conducting systematic safety evaluations for CAVs within existing traffic systems, without compromising realism. In response, our work leverages co-simulation strategies to propose a framework for validating CAV safety performance in a large-scale testing environment. Specifically, this study addresses the fundamental challenges in CAV safety validation by introducing a comprehensive evaluation framework within a vast 20×20 km^2 testing environment. This approach substantially extends beyond traditional small-scale testing methods. Our large-scale, high-fidelity simulation environment compensates for the limitations of real-world testing by enabling controlled, repeatable experiments across diverse scenarios, while still maintaining a close approximation to realistic driving conditions.

The remainder of this paper is structured as follows: Section 2 outlines the proposed testing framework. Section 3 elaborates on the credibility of the ADS and CAV functions. Section 4 presents a case study using the developed simulation platform and its associated analysis. Section 5 concludes the paper by summarizing and discussing the key findings.

2. Overview of the testing framework

This section offers an overview of the high-level (i.e., L4) CAV testing framework, emphasizing a co-simulation-based approach that integrates traffic microsimulation with CARLA. Specifically, this work introduces a comprehensive methodological framework for the systematic safety assessment of CAVs across testing tracks ranging from small (1×2 km^2) to large (20×20 km^2) scales within a single testing environment. The framework is divided into three main components (Fig. 1): the creation of the testing environment, the simulation of ADS, and the development of safety measurement metrics. This section primarily focuses on the development of the testing environment and the design of measurement metrics, while detailed information regarding vehicular autonomy and connectivity are covered in Section 3.

2.1. The establishment of the testing environment

In this study, the testing environment is composed of static and dynamic elements. Static components include terrain, scenery, infrastructure, road structures and layouts, while dynamic elements encompass variations in weather and lighting conditions, traffic flow, and operational driving instructions. Specifically, this work utilizes CARLA platform version 0.9.15, incorporating elements from "Town 12" and "Town 13" to create a 20×20 km^2 static environment. This environment replicates a diverse range of road geometries found in high-rise, residential, and rural areas, mirroring the complex urban layouts typical of large North American cities (CARLA Simulator, 2023). As depicted in Fig. 2, the process of establishing a static environment involves three primary steps: terrain adjustment, layout determination, and road network construction. Following these steps, the weather and lighting components are integrated using the built-in weather system of the Unreal Engine. This system has been validated in literature for its accuracy in replicating real-world conditions (Sørensen and Jørgensen, 2022).

Traffic dynamics, a crucial aspect of the dynamic elements, are generated using VISSIM. This traffic microsimulation software acts as a dedicated terminal for managing traffic volume, road user composition, traffic rules, and traffic signals. Technically, the road layout is modelled in VISSIM to match the static environment, and traffic flow rates are set between 1,262 vehicles per hour and 21,000 vehicles per hour, based on the Annual Average Daily Traffic (AADT) data for larger cities in North America as reported by the Federal Highway Administration (2024). A critical element in generating traffic within this framework is the application of the Wiedemann 99 (W99) behavioral model (Wiedemann and Reiter, 1992), which outlines ten parameters that dictate the operational logics of the simulated road users. This model intricately simulates vehicular behavior by incorporating both physical and psychological driving factors, such as stationary and dynamic following distances, perception thresholds, and acceleration behaviors.

In this study, VISSIM road users are pre-configured to exhibit a range of driving styles from very comfort-oriented to neutral, and extending to very aggressive, including illegal driving behaviors. Specific adjustments to the behavioral model parameters that influence acceleration and deceleration patterns, speed curves, and behaviors are applied to simulated road users. Empirical analyses suggest that road users' intentional rule-breaking behavior involves both conscious and unconscious decisions to violate traffic rules, with offenders committing a wide range of traffic offenses. Approximately 10–33 % of road users are intentional rule-breakers (Corbett and Simon, 1992; Freeman and Rakotonirainy, 2015; Devalla, 2018). Additionally, Wirth et al. (2018) noted that although rule-breaking behavior is shaped by a complex interplay of psychological, demographic, and situational factors, it is typically rational and calculated rather than random. Such behavior can be attributed to both the psychological characteristics of road users and the physical characteristics of the road system. To this end, we assume a proportion of 10 % rule-breaker in the simulated large-scale testing environment, and validate the rule-breaker ratio RB via pre-testing in the simulation environment, fundamentally based on the following equation:

$$RB = \frac{\sum (V_i \times F_i \times L_i)}{N_o} \tag{1}$$

where V_i, F_i, and L_i denote the severity weight, frequency, and location for violation type i, respectively. N_o denotes the total observations during the pre-testing phase. Within the developed 20×20 km^2 testing environment, the weighted methodology accounts for spatial variations in road users' behavior, temporal pattens, violation types, and location-

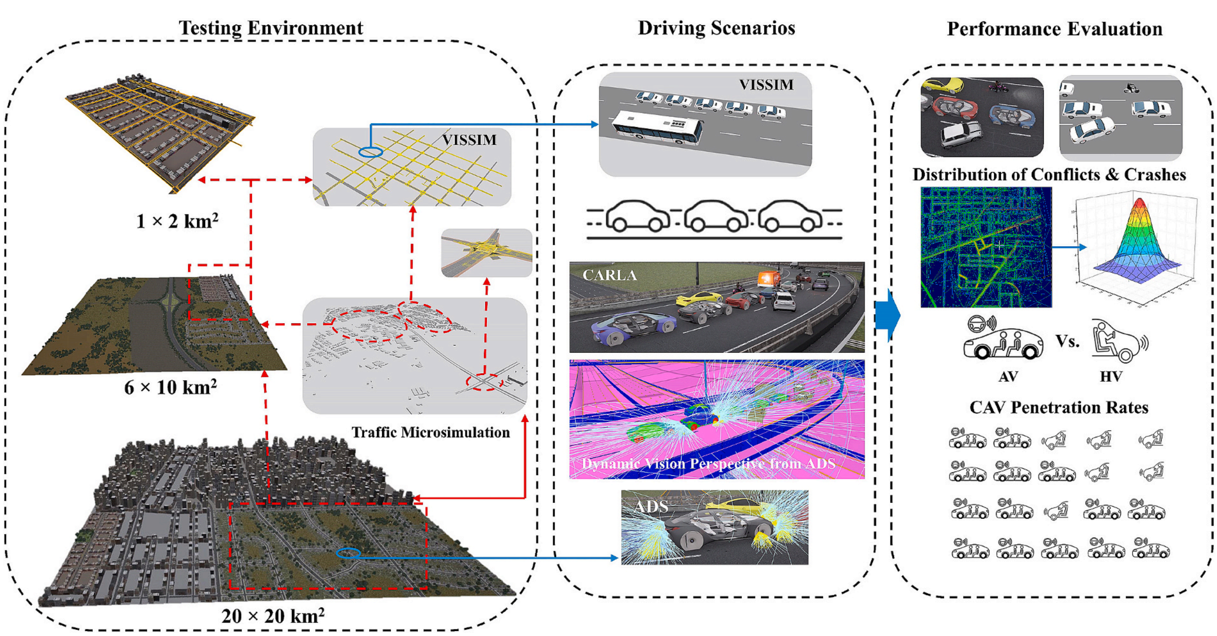

Fig. 1. Overview of the proposed methodological framework for CAV performance evaluation: This framework centers on evaluating traffic conflicts and crashes, comparing human-driven vehicles with autonomous vehicles, and examining the impact of CAV penetration rates. These elements correspond to specific driving scenarios within the testing environment, which is constructed using a co-simulation strategy that integrates VISSIM with the CARLA simulation platform.

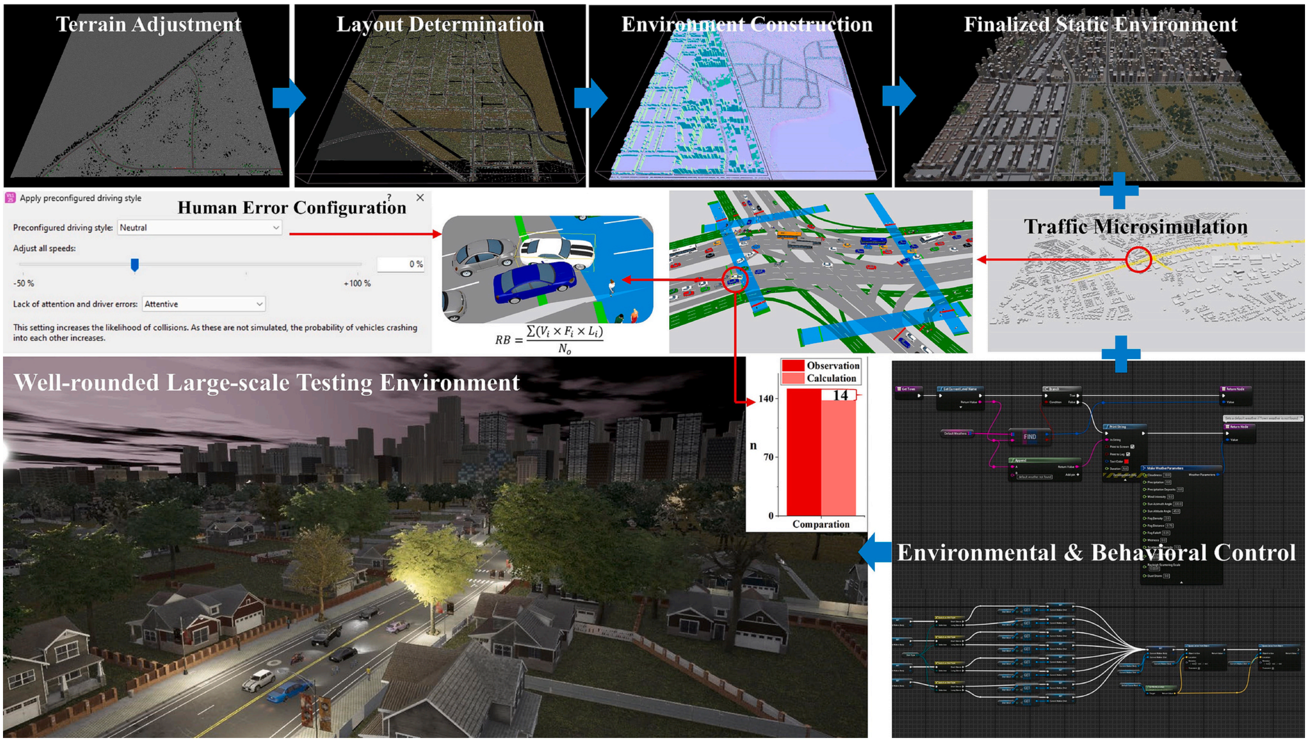

Fig. 2. Overview of the establishment of the testing environment: beginning with terrain adjustments in CARLA and progressing to the completion of the static environment. This is followed by the generation of traffic dynamics using micro-simulation, which includes modelling traffic flow and road user behavior. The process culminates with dynamic environmental and behavioral control, achieved through a co-simulation strategy.

specific factors.

Based on the 95 % confidence level to sampling the rule-breaking cases using the following equation:

$$n = \frac{\frac{Z^2pq}{E^2}}{1 + \left(\frac{Z^2pq}{E^2N_o}\right)} \quad (2)$$

where the Z-score value is 1.96, p is the expected rule-breaker proportion, q is the proportion of rule-followers, E represents marginal error.

Theoretically, in a large-scale road system with a total area of 400 km^2, a general assumption of standard lane capacity could be 1,800 to 2,000 vehicles/hours/lane, with the average number of 2 to 3 lanes per direction. Based on the network utilization factor of 0.3 (i.e., 30 % of roads at capacity during peak periods), the total observation could be calculated:

$$N_o = l_T \times l_n \times Q \times \mu \quad (3)$$

where l_T is the total road length, ln denotes the average lane numbers, Q is flow rate, and μ denotes the utilization factor.

By adjusting the proportion of human error (p) to 10 %, we observed a slight difference of 14 cases in n when comparing results from the simulation with calculated outcomes, as depicted in Fig. 2. Thus, we claim that the 10 % proportion is reasonable and can be validated through systematic observation and data collection. Such targeted adjustments ensure that the simulation accurately reflects the variability and unpredictability observed in actual driving behaviors. Utilizing such configurations allows for the replication of more realistic driving patterns and integrates human driving errors into the simulated environment, enhancing the reliability of the simulation (Astarita and Giofré, 2019; Siebke et al., 2022).

The concept of co-simulation in this study is built upon the Component Object Model (COM) interface, which facilitates the integration of VISSIM-generated traffic dynamics into the static environment. This interface enables VISSIM objects to be accessed and executed from the CARLA platform in real-time, a method validated in numerous prior studies (Al-Msari et al., 2024). Additionally, we have incorporated the Unreal Engine Navigation System, driven by the Reciprocal Velocity Obstacles (RVO) algorithm (Sun et al., 2022), and a multi-layer behavior modelling strategy (Colledanchise and Natale, 2021) into all VISSIM-generated road users. This incorporation allows for the accurate reproduction of speed variations in VISSIM-generated road users under varying weather conditions, reflecting different road surface conditions. For more detailed information on the development of this co-simulation framework, readers are referred to our previous works (Xu et al., 2022, 2023, 2024a).

To better reflect the realism driving environment, the behavior of Adaptive Signal Control Technology (ASCT) systems is incorporated into the simulation using Python scripts in the form of client–server mode. Specifically, all traffic lights within the large-scale testing environment are retrieved and working collaboratively under adaptive control logic. Fundamentally, the traffic rate (q) of a road segment is monitored in real-time during the simulation, and the cycle length (C) of traffic lights can be estimated using Webster's formula. Thus, the proportion of green time (G) for each phase can be determined based on the traffic volume serving each phase relative to the total intersection traffic volume:

$$C = \frac{1.5L + 5}{1 - \sum_{i-1}^{n} \frac{q_i}{s_i}} \quad (4)$$

$$G = \frac{\frac{q_i}{s_i}}{\sum_{i-1}^{n} \frac{q_i}{s_i}} \times C \quad (5)$$

where q_i is the flow rate for phase i, s_i is the saturation flow rate for phase i, L is the total loss time.

As the critical flow ratio $\sum_{i-1}^{n} \frac{q_i}{s_i}$ approaches or exceeds 1, the demand during the green phase meets or surpasses the capacity, necessitating adjustments in signal timing. In this case, the adaptive control logic determines the appropriate phase for each traffic light based on real-time traffic monitoring in CARLA, which involves analyzing the current traffic conditions such as vehicles queues, traffic density, time since last green, etc. Such integration facilitates the modification of signal timings to adapt to varying traffic conditions, enhancing the realism of the simulated traffic environment and more accurately reflecting real-world traffic control practices.

Fig. 3 offers a detailed example of the implementation of ASCT within the simulation environment, showcasing 8 signalized intersections that are adaptively controlled. CARLA actively monitors traffic conditions for each segment and updates traffic light phases according to a construction script that outlines the control logic. For instance, as depicted in Fig. 3, the green time (G) at two consecutive intersections is increased during light traffic flow conditions. This adaptive control function is applied consistently across all signalized intersections within the large-scale simulated testing environment, operating in a continuous loop throughout the simulation. In addition to communication between intersections, this control logic also integrates pedestrian crossings and emergency vehicle priority, enhancing the realism and functionality of the simulated traffic management system.

2.2. Safety measurement metric

This work concentrates on the safety assessment of high-level CAVs within the established large-scale testing environment. The primary objective is to systematically quantify traffic conflicts and crashes. The safety measurement metric employed defines quantitative indices, such as collision probability and accident rates, using surrogate safety assessment. Based on the observed safety performance, the analysis then addresses the importance of potential conflict-contributing factors, employing a spatial–temporal framework to structure these considerations.

Fundamentally, the dynamic driving tasks (DDT) performed by CAVs are within the given operational design domain (ODD) in this testing framework, and the controlled variables are defined in the following metric:

$$\mathbb{X} = \begin{bmatrix} R_1, R_2, \cdots, R_i \\ L_1, L_2, \cdots, L_i \\ W_1, W_2, \cdots, W_i \\ \vdots\vdots\vdots \\ BV_1, BV_2, \cdots, BV_i \end{bmatrix}, \mathbb{X} \in ODD \tag{6}$$

where R, L, W, and BV denote road type, lane number, weather condition, and background vehicle status, respectively. Note that the $\mathbb{X}$ includes a vast number of combinations and is corresponding to specific traffic event.

In this study, surrogate safety measurements are employed to quantify traffic events, focusing on key safety indicators such as Time-to-Collision (TTC) and Post-Encroachment Time (PET). These measurements encompass a broad spectrum of traffic conflicts, including vehicular, vehicle-bicycle, vehicle–pedestrian, and vehicle-infrastructure interactions. TTC measures the time remaining until a collision could occur between two road users, based on their current positions and velocities. PET, on the other hand, calculates the temporal difference between the moments when two road users pass through the same conflict point. The mathematical formulas used to calculate these indicators are presented below:

$$TTC_i = \frac{X_{i-1}(t) - X_i(t) - l_i}{\dot{X}_i(t) - \dot{X}_{i-1}(t)} \quad \forall \dot{X}_i(t) > \dot{X}_{i-1}(t) \tag{7}$$

$$PET_i = t_i - t_{i-1} \tag{8}$$

where the TTC between the road user i at instant t and its leading road user $i-1$ is estimated by their speeds $\dot{X}_i$ and vehicle lengths l_i; the PET is determined by the difference between the instantaneous time of the first road user leaving the conflict point t_{i-1} and the time that second road user reaching the conflict point t_i. Given the ability of high-level ADS to securely maintain close proximity during operation, the critical safety thresholds for TTC and PET have been set at 1 s. These thresholds are lower than what are typically used for conventional vehicles (CVs) (i.e., 1.5 s and 3 s), reflecting the enhanced response capabilities of ADS. This adjustment aligns with the results from previous research (Alozi and Hussein, 2022; Xu et al., 2022).

Statistically, a crash is considered event when the safety indicator—either TTC or PET—reaches 0 or approaches it very closely. To comprehensively assess the risk associated with such extreme events, this study employs a multi-type Bayesian hierarchical model (MBHM) as detailed by Fu et al. (2021). This model effectively incorporates the impacts of controlled variables $x_i \in \mathbb{X}$ and accommodates potential correlations among traffic events within a spatial–temporal structure. The underlying methodology of this model is rooted in extreme value analysis, utilizing the Generalized Extreme Value (GEV) distribution (Beirlant et al., 2006). This approach allows for a robust assessment of the most severe outcomes predicted by the safety indicators:

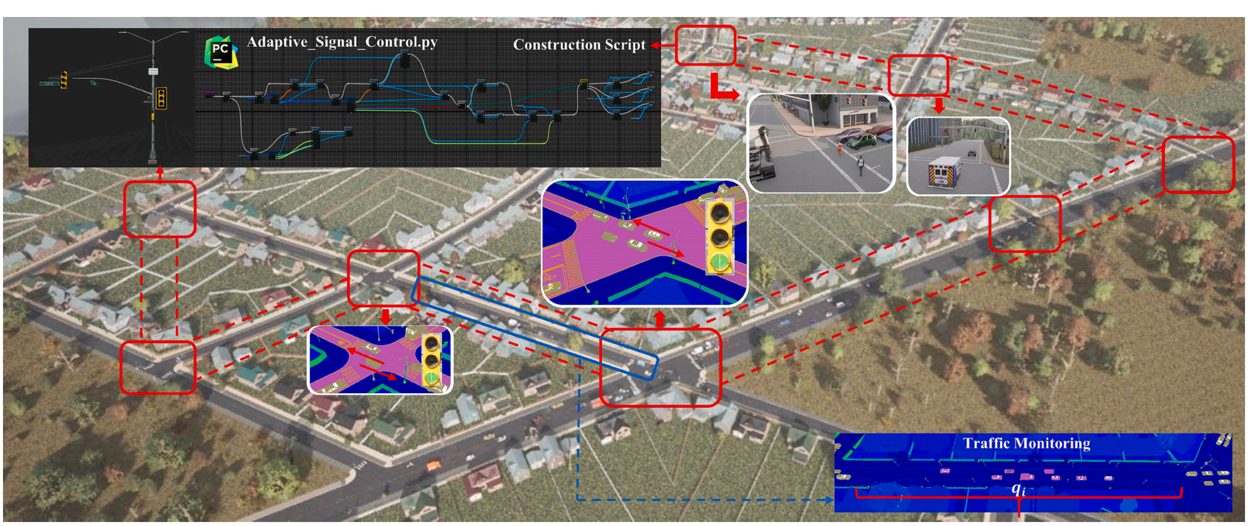

Fig. 3. An illustrative example of the implementation of Adaptive Signal Control Technology (ASCT) within the simulation environment.

$$G_i(o_i) = \exp\{-\left[1+\frac{\xi_i(o_i-\mu_i)}{\sigma_i}\right]^{\frac{-1}{\xi_i}}\} \quad (9)$$

$$\forall\, o_i \in R, \quad 1+\frac{\xi_i(o_i-\mu_i)}{\sigma_i} > 0 \quad (10)$$

where G denotes a n-variate distribution function, in which n types of traffic conflicts are considered and measured with conflict indicators. This function is with non-degenerate margins. o is a vector of independent random observations, ξ_i is the shape parameter, μ_i is the location parameter, and σ_i is the scale parameter.

In the modelled large-scale testing environment, each lane is uniquely identifiable, allowing for detailed analysis of traffic dynamics. For example, the traffic event o_{ijk} represents the i th recorded traffic conflict with the j th type at the site k. These events (including the numbers, types, and locations of conflicts/crashes) are captured by the simulation platform and serve as inputs to the MBHM. This setup facilitates the estimation of the influence of potential contributing factors to traffic conflicts using the multi-type GEV distribution:

$$G(o_{i1k} < o_1, \cdots, o_{ink} < o_n) = \exp\{-\left[\sum_{j=1}^{n}\left(1+\frac{\xi_{ijk}\left(o_j-\mu_{ijk}\right)}{\exp(\varphi_{ijk})}\right)^{\frac{-1}{\xi_{ijk}}}\right]^{\alpha}\} \quad (11)$$

where the scale parameter σ_i is further considered as $\exp(\varphi_i)$, and the dependence parameter $\alpha \in (0,1)$ is assigned as a uniform distribution to balance the shape parameter ξ_i. This statistical approach helps in understanding how various factors interact within the environment to influence safety outcomes.

2.3. System validation

To ensure that the simulated environment accurately reproduces traffic conditions, the microsimulation model underwent a comprehensive validation process. Simulated outputs were compared against real-world data collected using the TomTom Traffic API (Tomtom Developers, 2022), which has been well-validated for accurate traffic data collection (Bauer et al., 2019; Ranjane et al., 2023). This rigorous validation helped confirm the reliability and accuracy of the simulation in reflecting actual traffic behaviors and conditions.

A representative 2.3 × 1.6 km^2 area in Baltimore, MD, USA, with a speed limit of 50 km/h, was selected as the study area for system validation. This area, where six incidents were recorded at a signalized intersection, is not only a black spot for traffic incidents but also effectively represents the driving patterns typical of residential and commercial areas in North America. The validation approach incorporated multiple metrics that demonstrate a strong alignment between simulated and real-world conditions. As illustrated in Fig. 4, panels (a) and (b) compare the actual road layout of the selected area for validation with a similar layout generated in the simulation, showing alignment in terms of road design and scale. Traffic patterns are indicated by colors on the road segments: red signifies congested areas with an average speed of 25 km/h, orange represents less congested roads with an average speed of 36 km/h, and green indicates free-flowing traffic with speeds of 48 km/h. The simulated traffic environment accurately reproduced the observed patterns, and the traffic flow validation presented in Fig. 4c shows a decent correlation between the simulated and actual counted traffic flows, with an R^2 value of 0.98. Regarding traffic conditions, we validated the average speeds across these three distinct flow states: free flow, less-congested, and congested conditions. The simulation closely matched real-world measurements, with comparable speed distributions (Fig. 4d):

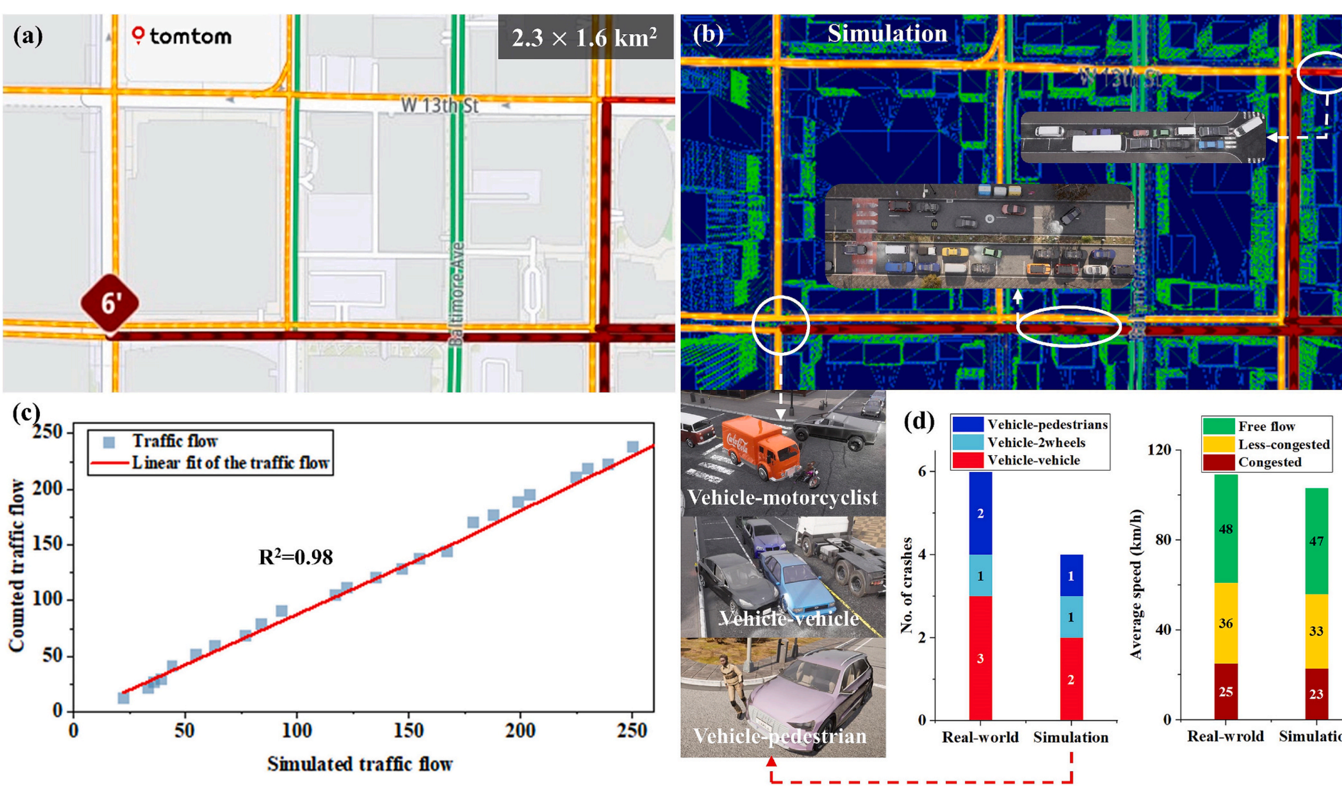

Fig. 4. Validation of the simulation model: (a) an overview of the TomTom map displaying the typical study area for validation; (b) simulation environment with identified road layout with representative points being demonstrated; (c) correlation between simulated and counted traffic flows; (d) comparative analysis of crash patterns and average speeds between real-world data and simulation results.

- Free flow: 48 km/h (real-world) vs. 47 km/h (simulated)
- Less-congested: 36 km/h (real-world) vs. 33 km/h (simulated)
- Congested: 25 km/h (real-world) vs. 23 km/h (simulated)

This high coefficient of determination and close alignment confirm that the simulation accurately reproduces real-world traffic flow rates, demonstrating that the simulation reliably captures the fundamental traffic dynamics of the study area.

Regarding crash patterns and safety validation, we compared the typology of conflicts (Fig. 4d). The real-world data showed 6 crashes (3 vehicle-vehicle, 1 vehicle-2wheels, and 2 vehicle–pedestrian) at the intersection, while our simulation predicted 4 crashes (2 vehicle-vehicle, 1 vehicle-2wheels, and 1 vehicle–pedestrian) at the same place. This close alignment in both total numbers and distribution of crash types validates the capability of the proposed simulation strategy to represent safety–critical scenarios.

3. The development of intelligent CAV function

This section outlines the development of the ADS intelligence using the co-simulation platform, which is crucial for establishing the reliability of tests conducted on CAVs. Specifically, it covers the calibration of ADS within the established large-scale testing environment, illustrating how the ADS processes the entire testing area as its ODD to safely perform DDTs. This process demonstrates the ADS's capability to adapt and respond to the comprehensive and complex scenarios presented in the testing environment.

3.1. Vehicular autonomy

Thanks to the vehicle pre-configuration on the CARLA platform, the kinematic parameters are finely adjusted to reflect real vehicle physics, enhancing realism. This adjustment includes accurate simulations of tire force, vehicle chassis feedback, and responses to frictional forces (Dosovitskiy et al., 2017). Such features are standard and transferable across all scenes within the CARLA platform. Furthermore, the ADS control model in CARLA is meticulously refined through an extensive training set comprising over 10 million iterations, effectively simulating the functionality of a multi-sensory system. Consequently, CARLA offers an autopilot mode that simulates high-level autonomous driving optimally across all integrated maps (e.g., Town 1 to Town 10) without additional model integration. However, integrating traffic micro-simulation disrupts these optimal environmental conditions by eliminating fixed spawn points and trajectories for non-player road users. The established 20×20 km^2 testing environment also significantly diverges from CARLA's original maps, which introduces new ODD boundaries. Thus, a recalibration is required for the ADS control model.

In this study, the ADS control model was recalibrated within the established testing environment. The objective was to enhance the interaction between the ego vehicle and its surroundings through a continuous sequence of observations, actions, and rewards. The aim for the ego vehicle was to select actions that maximize cumulative future rewards, thereby effectively treating the established environment as its ODD. To achieve this, we employed the deep reinforcement learning architecture initially proposed by Mnih et al. (2015). This architecture utilizes a deep convolutional neural network (CNN) to approximate the optimal action-value function. The network is trained to predict the maximum sum of discounted future rewards achievable from each state-action pair, facilitating the ego vehicle's decision-making process to optimize its path and interactions within the ODD. This deep learning approach enables the ADS to learn complex behaviors based on high-dimensional sensory inputs, mirroring human-like driving capabilities within the simulated environment. Fundamentally, it is built up the following action-value function:

$$Q^{*}(s,a) = \max_{\pi}\mathbb{E}\left[r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \ldots | s_t = s, a_t = a, \pi\right] \tag{12}$$

here, in the pre-defined testing environment $\mathbb{E}$, r_t denotes the reward received at time-step r_t, and γ is the discount factor, which weighs the importance of future rewards. The behavior policy $\pi = P(a|s)$ defines the probability of taking action a given the state s.

Based on this architecture, the ego vehicle could persistently execute actions in each subsequent state s_{t+1} to obtain rewards $r(s_t, a_t)$ within the established large-scale testing environment. The behavior is dictated by policy π, which encapsulates the fundamental traffic rules and interactions within a stochastic setting. To reinforce adherence to this policy, Q-learning updates are employed. The loss function used in the Q-learning update at iteration i is as follows:

$$L_i(\theta_i) = \mathbb{E}_{(s,a,r,s')\sim U(D)}\left[\left(r + \gamma \max_{a'} Q(s', a'; \theta_i^-) - Q(s, a; \theta_i)\right)^2\right] \tag{13}$$

where the network parameter θ_i represents the weights and biases of the neural network at iteration i, γ is the discount factor and θ_i is the network parameters used to compute the target at iteration i, which is periodically updated with the weights of the main Q-network to stabilize learning. The target network helps to mitigate the issues of moving targets in training that can arise from using a constantly updating policy to both generate and evaluate actions.

Employing this training strategy, the ADS achieved effective and safe operation within 10,000 iterations. Fig. 5 presents an overview of the ADS recalibration results. Fig. 5a examines the behavior of the ADS and its interactions with the environment and other road users. Red bars indicate the lateral position of the ego vehicle across iterations. The varying heights of the bars illustrate adjustments in the vehicle's lateral position, showing a progression from a fluctuation of 4 m to a stable position of less than 1 m from the center of the lane after 7000 iterations. Blue bars display the accident rate for the ego vehicle over the same iterations, overlaid on the same graph for direct comparison. This shows that safety performance could be assured after more than 8000 iterations, as evidenced by a reduction in accident rates from over 60 % to less than 1 %. In Fig. 5b, the bar graph displays the frequency of different types of accidents per 100 km across a series of iterations, assessing the frequency with which the ADS was involved in accidents with other vehicles or road infrastructure. This panel highlights that vehicular interaction was the primary cause of safety–critical events during the training phase. Compared to the benchmark performance of CARLA's default autopilot function, the fine-tuned ADS in the established testing environment surpasses CARLA's default autopilot in terms of vehicle position control and accident avoidance. Fig. 5c depicts a typical roadway scene during a training session in rainy conditions, captured from vehicle sensors including the original camera, optical sensor, dynamic vision sensor, depth camera, and LiDAR. This sub-figure visualizes the training process of the ADS, demonstrating its reliance on multiple sensor inputs for real-time navigation decisions as developed during recalibration.

3.2. Vehicular connectivity

Working upon the recalibrated ADS control model, vehicular connectivity was enhanced by incorporating a utility function that determines platooning decisions based on travel cost (such as energy consumption, reaction to situations, and motion efficiency) and travel time to a specific destination. This function theoretically calculates the perceived utility of selecting an alternative route to optimize overall throughput:

$$U_k(T) = -\vartheta C_k[\nu(T)] + \varepsilon_k(\nu), \forall k \in O_s(T) \tag{14}$$

where C_k is a function of the deterministic component that captures the

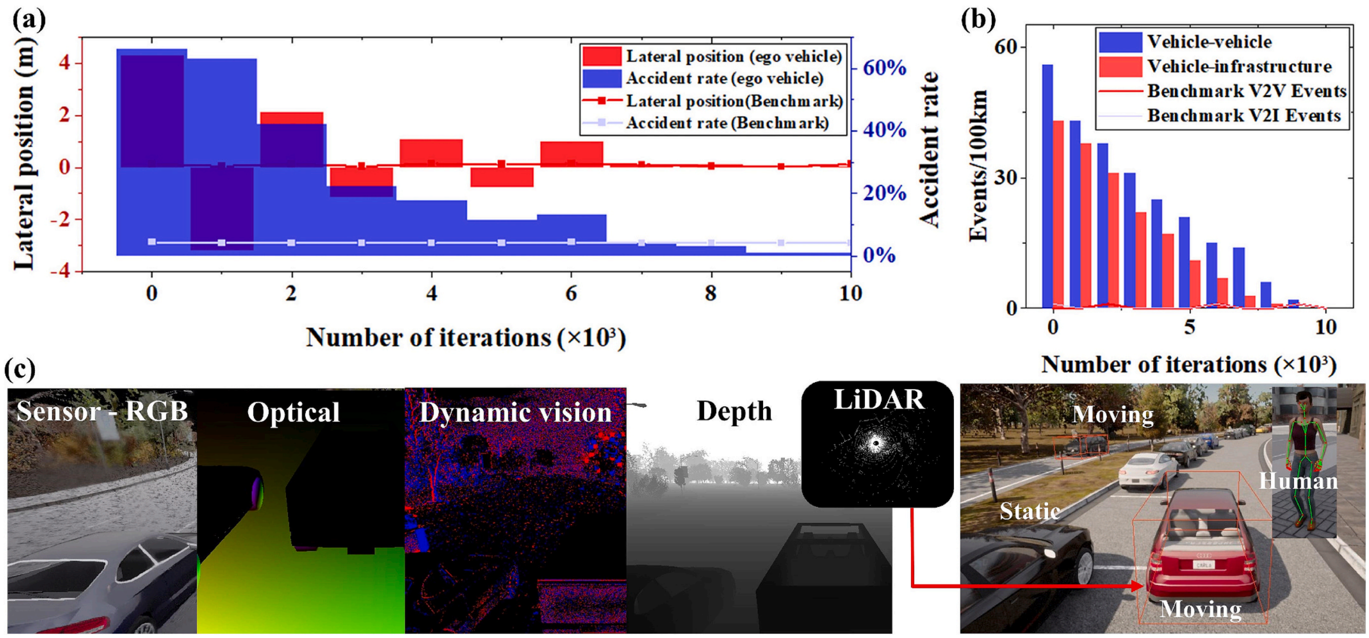

Fig. 5. ADS recalibration in the established large-scale testing environment: a multi-faceted evaluation of ADS over iterations. Panel (a) illustrates changes in lateral position and accident rates for an ego vehicle versus CARLA's benchmarks. Panel (b) categorizes accident rates by interaction type, comparing vehicle-to-vehicle and vehicle-to-infrastructure incidents against CARLA's benchmarks. Panel (c) showcases a variety of sensor outputs along with object classification in an urban driving scenario, highlighting the integration of multiple sensor data for enhanced vehicle perception during the training.

known or certain outcomes associated with a particular choice or action; $v(T)$ is the vector of the values at time t of the variables on which this utility depends; $\varepsilon_k(v)$ is an additive random error, representing the perception error due to the lack of perfect information; ϑ is a positive variable to balance the negative utility. All potential alternatives fall within $O_s(T)$, which represents the set of all available paths from the current position to the destination s at the time interval T.

Additionally, the decision to connect or disconnect largely depends on the route choice and any delays in information exchange. The theoretical foundation for the platooning decision is outlined below:

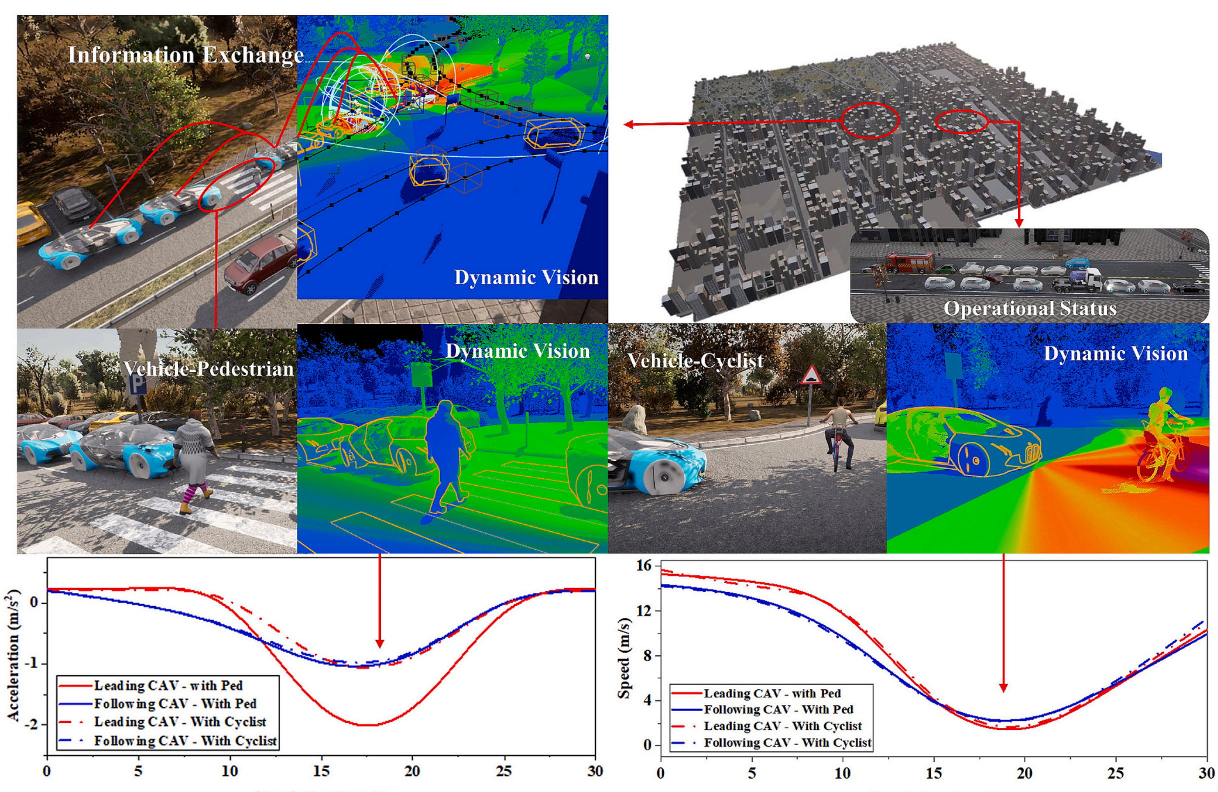

Fig. 6. The operation of CAVs within the established large-scale testing environment: an informative visualization encapsulates the technological aspects of CAV operations, focusing on CAV-involved interactions that highlight dynamics of acceleration and speed variations.

$$T_a^{n,s} = (1-\sigma)T_a^{n-1,s} + \sigma TO_a^{n,s} \tag{15}$$

where $T_a^{n,s}$ represents the expected travel time on edge a during period s in the n th iteration, $TO_a^{n,s}$ is the measured travel time as a baseline, σ denotes a smoothing factor. As a result, every single CAV could decide whether to form a platoon according to the best route selection for each O-D pair dynamically.

Message transmission delay, as a tangible issue in real-world scenarios (Malik et al., 2020), is also accounted for in the simulation. However, it is reasonable to hypothesize that high-level CAVs can compensate for such delays to a certain degree. To address this, we incorporate a vehicle motion optimization function designed to maintain a roughly constant acceleration for all CAVs within or approaching a fleet during the delay period. By applying the fundamental equations of motion and assuming constant acceleration for all vehicles during the delay, CAVs can accurately estimate the speeds and positions of other vehicles nearby:

$$v_p^{est} = \left[v_p + a_p T_D'\right]_{t-T_D'} \tag{16}$$

$$x_p^{est} = \left[x_p + \frac{a_p T_c'^2}{2} + v_p T_c'\right]_{t-T_c'} \tag{17}$$

where v_p, x_p, and a_p denote the speed, position, and acceleration of a preceding CAV delayed by time T_c', respectively. The underlying car-following model (W99) straightforwardly derives the actual parameters characterizing vehicle physics, considering the subject vehicle's current speed and position as well as the estimated speeds and positions of all other connected vehicles in its vicinity.

Based on these strategies, Fig. 6 depicts the operation of CAVs within the established large-scale testing environment. It specifically illustrates typical road user interactions, demonstrating how CAV perception systems process various environmental elements. This includes vehicle-to-vehicle dynamics as well as interactions with cyclists and pedestrians, each represented in distinct visual layers to highlight sensor readings and situational awareness. Similarly to individual ADS, CAVs can achieve a stable operational state by continuously detecting and predicting the behavior of road users, thereby enabling all activated vehicles to form a fleet. As shown in Fig. 6, within the CAV fleet, every vehicle, regardless of type, can maintain a stable acceleration rate; thanks to vehicular connectivity strategies, trailing vehicles can achieve smooth acceleration variations of less than 1 m/s^2, while maintaining a minimal speed difference of less than 2 m/s compared to the lead vehicle in most cases. For more detailed information, readers are referring to our previous works (Xu et al., 2024a,b, 2024).

4. Case study

This section systematically evaluates the safety performance of CAV operations within the large-scale testing environment. Employing a comparative approach, the case study systematically measures traffic conflicts and crashes, considering the effects of all controlled variables as outlined in Eq. (6), including terrain and road type, traffic flow conditions, weather conditions, CAV penetration rates, etc.

4.1. Case study design

To validate CAV performance within the established 20 × 20 km^2 testing environment, the simulation was conducted in a comparative manner. Based on a cycle time of 3600 s, a total of 202 simulation runs were completed, spanning CAV penetration rates from 0 to 100 % under both low and high traffic flow conditions. Weather conditions during the simulation were randomly set, encompassing controlled variations of rain, snow, and fog—ranging from light to heavy—as well as clear weather conditions during both daytime and nighttime. Although the simulation included the entire testing environment, it was divided into four distinct areas based on terrain, road types, infrastructure designs, and traffic flow rates for detailed evaluation. As shown in Fig. 7, these are the high-density residential area (A1), central business district (CBD) & medium-density residential area (A2), medium-density area (A3), and rural area (A4). Each area features unique road layouts, infrastructure, and traffic conditions, yet they are seamlessly connected. Table 1 summarizes the characteristics of each area.

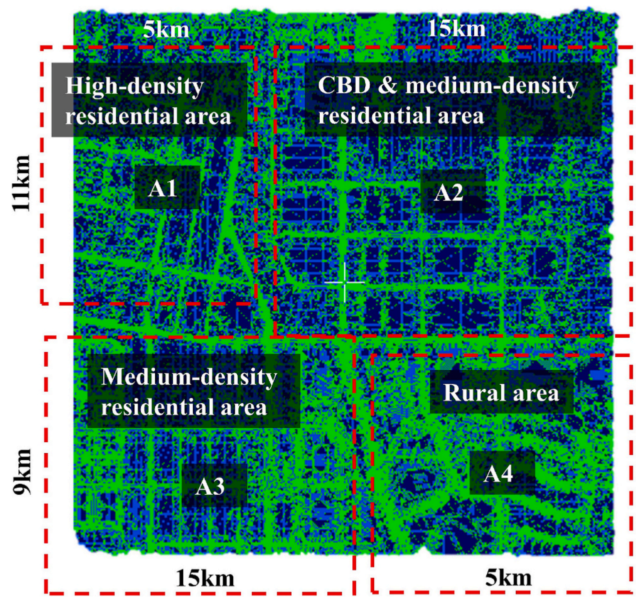

Fig. 7. The layout of the established 20 × 20 km^2 testing track.

The division of the study area could assist the safety evaluation of CAV operations comprehensively because of the following key features.

- *Traffic flow and congestion impacts:* Each area exhibits distinct traffic patterns and volumes. According to the Federal Highway Administration (2024), high-density and central business district (CBD) areas are expected to experience significant congestion, frequent stop-and-go conditions, and complex intersection behaviors. These conditions present a challenge for CAVs in managing dense urban traffic, with traffic flow varying from 304 veh/h to 8941 veh/h. Conversely, medium and low-density areas experience lighter traffic volumes, ranging from 284 veh/h to 4033 veh/h, and may present different challenges such as higher speeds and fewer interactions with other vehicles. In rural areas, where traffic flow varies from 197 veh/h to 3706 veh/h, roads typically feature fewer lanes but higher speed limits compared to urban settings.
- *Traffic efficiency and road user interaction:* Different areas provide valuable insights into how safely and efficiently CAVs operate across varying environments. For instance, high-density and CBD areas, where interactions with pedestrians, cyclists, and motorcyclists are more frequent, are crucial for testing the safety performance of CAVs in complex environments. In contrast, rural areas might concentrate more on scenarios involving vehicle interactions and navigating roads that are poorly lit, offering a different set of challenges and testing conditions for CAVs. These varying conditions help to comprehensively evaluate the adaptability and safety protocols of CAVs in diverse traffic situations.
- *Behavioral adaptation:* Observing how CAVs adapt to different areas can be instrumental in identifying ODD boundaries. For example, the transition from a high-density urban area to a rural setting requires CAVs to adjust to varying speed limits, traffic volumes, and roadway characteristics. This adjustment process is crucial for pinpointing potential limitations in CAV operations, such as their ability to

Table 1
Characteristics of each area.

	A1: High-density residential area	A2: CBD & medium-density residential area	A3: Medium-density residential area	A4: Rural area
Scale	11×5 km^2	15×13 km^2	15×9 km^2	7×5 km^2
No. of lanes	1 to 3	2 to 4	1 to 3	1 to 2
Speed limits	40 – 50 km/h	50 – 80 km/h	40 – 60 km/h	80 – 120 km/h
Roundabouts	8	7	6	5
Signalized intersections	9	18	15	4
Traffic flow conditions	304 – 4320 veh/h	477 – 8941 veh/h	284 – 4033 veh/h	197 – 3706 veh/h

handle changes in environmental conditions and traffic complexity, thereby highlighting the critical areas for further development and refinement in autonomous driving technologies.

4.2. Safety evaluation

Over 100,000 trajectories were generated during the simulation and stored in.trj files. These files were processed using the surrogate safety assessment model, which provided outputs that included the distributions of TTC and PET, number of crashes, and the probability of conflicts across the 202 simulation runs.

A total of 227 collisions were recorded during the evaluation period, with each simulation run experiencing at least one collision, irrespective of environmental conditions. Fig. 8 presents the results of a comparative analysis of the impact of CAVs on traffic safety. Fig. 8a & 8b illustrate the evolution of average TTC and PET as CAV penetration increases from 0 % to 100 %, benchmarked against scenarios with no CAVs where TTC and PET thresholds are set at 1.5 s and 3 s, respectively. The data indicate a general trend toward statistically unsafe conditions with increasing CAV penetration, characterized by reduced TTC and PET across all regions. For instance, the average TTC approached the 1.5-second threshold in conventional driving scenarios once CAV penetration reached 70 %, despite variations in road infrastructure, speed limits, and traffic flow conditions. Conversely, average PET values were effectively maintained above the 3-second safety threshold during all types of road user interactions throughout the study. The stability observed in PET contrasts with the significant reduction in TTC, suggesting differential impacts on safety metrics with higher CAV integration concerning interactions with different road users.

While the vehicular interaction indicator (TTC) neared or even below the safety threshold (Fig. 8a), the observed crash data (depicted in Fig. 8c & 8d) indicated a decrease in accident rates with higher CAV penetration. However, this safety benefit becomes significant only when CAV penetration exceeds 30 %, achieving a 51.16 % reduction in collisions under identical conditions. As CAV penetration increased, vehicle-to-vehicle collisions decreased markedly. This trend continued until CAV penetration reached 80 %, at which point a slight increase in the accident rate was observed. The rate continued to rise when CAV penetration reached 90 %, but then decreased slightly in a full-CAV scenario. The case study indicates that optimal safety performance was achieved with a 70 % penetration rate of CAVs, during which a total of six crashes were recorded, with three crashes occurring in each of the areas A1 and A2, respectively. Notably, there was a significant reduction of 86.05 % in accident rates (from 43 to 6 accidents) when the CAV penetration rate reached 70 %, compared to scenarios without CAVs. This supports the idea that increased automation can enhance safety. According to such observations, the safety benefits brought by implementing CAVs can be categorized into four phases (Fig. 9):

1. *Initial phase (0–20 % penetration):* At lower penetration rates, the safety improvements from each additional CAV are marginal due to the wide spacing between CAVs, meaning their safety influence

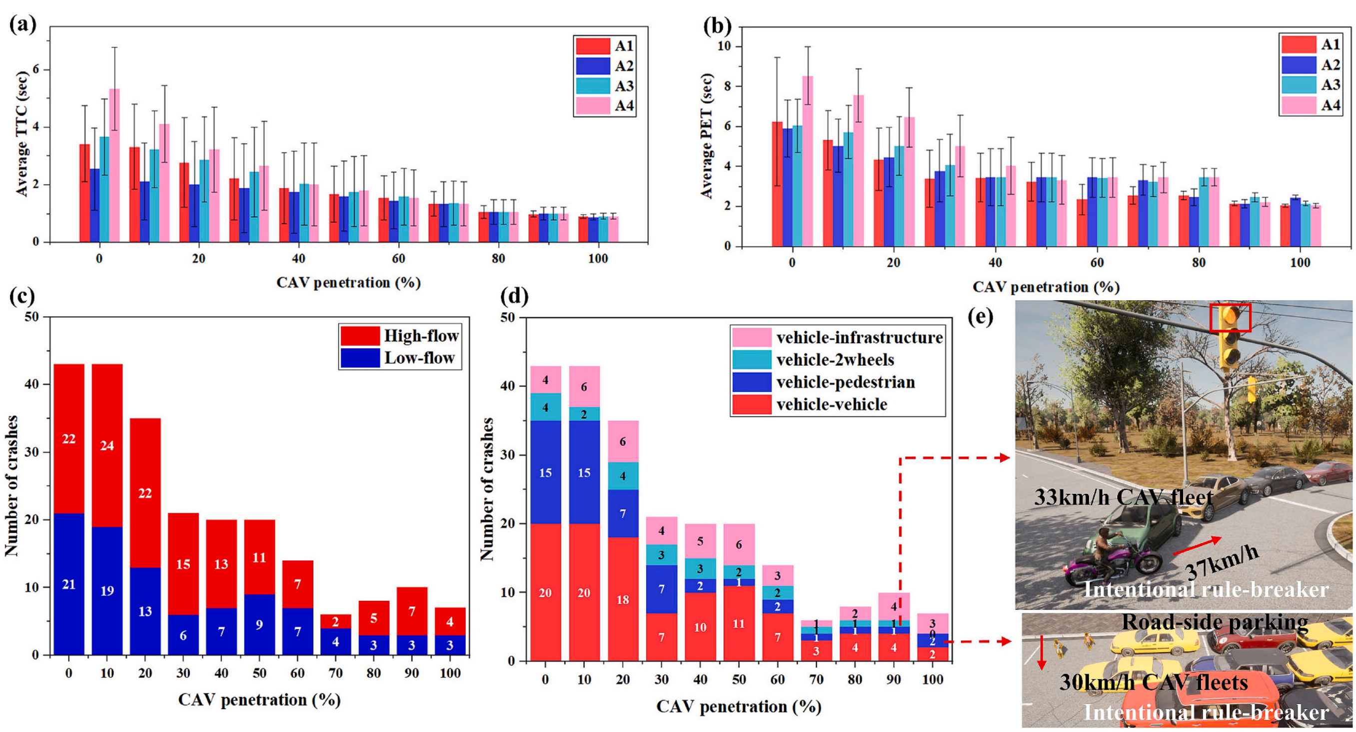

Fig. 8. Comparative analysis of CAV safety performance in the large-scale testing environment: (a-b) distributions of TTC and PET across various areas; (c-d) recorded crashes and the corresponding types of collisions observed; (e) representative incidents resulting from intentional rule-breaking behavior.

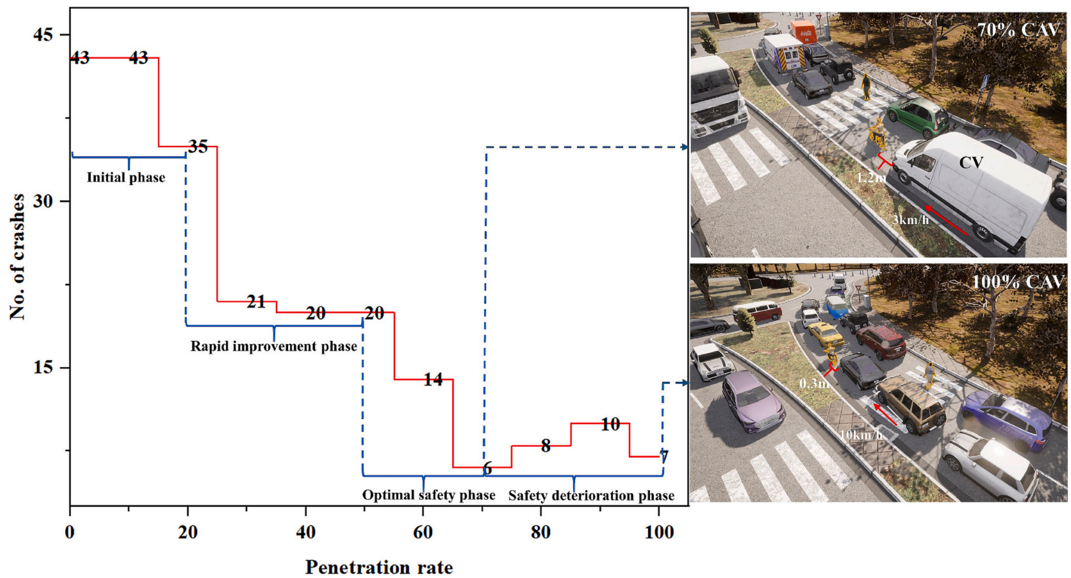

Fig. 9. Observed relationship between crash rates and CAV penetration rates, associated with illustrative examples for specific interactive scenarios at the optimal safety condition with a 70% CAV penetration rate and a full-CAV scenario.

zones do not significantly overlap. This is why the crash count remains high at 43 until about 15 % penetration. The first notable decrease in crashes, from 43 to 35, occurs at 20 % penetration, indicating that CAVs create zones of enhanced safety awareness that benefit both CAVs and CVs nearby. This suggests a threshold effect where a minimum penetration is needed to observe safety benefits.

2. *Rapid improvement phase (20–50 % penetration):* This phase sees the most significant safety improvements, with crashes sharply decreasing from 35 to 21. It represents the period of maximum marginal benefit per increase in penetration rate, suggesting effective interactions between CAVs and CVs.
3. *Optimal safety phase (50–70 % penetration):* The system reaches a temporary equilibrium in safety performance, and the benefits become more gradual. The optimal safety performance is achieved at 70 % penetration, with only 6 crashes identified. In this phase, the remaining 30 % of CVs likely benefit from being surrounded by CAVs while still maintaining sufficient diversity in driving behaviors.
4. *Safety deterioration phase (70–100 % penetration):* Beyond the optimal penetration rate, the homogenization of traffic behavior may lead to minor fluctuations and introduce new challenges, suggesting that maximum safety benefits occur at a specific penetration rate.

Furthermore, when concerning the interaction between vehicles and vulnerable road users, a decreased trend can also be found in vehicle–pedestrian and vehicle-2wheels (including both cyclists and motorcyclists) crashes as the CAV penetration increased (Fig. 8d). Vehicle-pedestrian crashes start at a higher level (15 crashes) and show a dramatic reduction when CAVs were introduced with 10–30 % penetration, dropping to 7 crashes, before steadily decreasing to 2 crashes at 60 % penetration and maintaining low levels thereafter. Vehicle-2wheels crashes begin at a lower level (4 crashes) and show a more gradual improvement pattern, fluctuating slightly in the early stages (dropping to 2 crashes at 10 % but rising to 4 crashes at 20 %), then stabilizing at 2–3 crashes in the middle range (30–60 %), and finally achieving optimal safety with 1 or fewer crashes at high penetration rates (>60 %). This demonstrates that while CAV technology effectively improves safety for both types of vulnerable road users, the benefits manifest differently, with pedestrian safety showing more dramatic early improvements compared to the more gradual enhancement of two-wheeler safety.

Nevertheless, there is a concerning trend in the interaction space between CAVs and vulnerable road users as the CAV penetration rate increased. Fig. 9 illustrates the observed results concerning the number of crashes in relation to CAV penetration rates. It also includes typical examples concerning the complex interactions between CAVs, CVs, pedestrians, and cyclists at the optimal condition with a 70 % CAV penetration rate and the scenario with full CAV penetration. At 70 % CAV penetration, there is a relatively comfortable safety buffer, with conventional vehicles maintaining a distance of 1.2 m from the cyclist while traveling at 3 km/h near crosswalks. In the 100 % CAV scenario, the safety buffer significantly narrows to just 0.3 m, while the operating speed increases substantially to 10 km/h. In this setup, CAVs pass pedestrians directly once their moving track is confirmed to be clear. This reduction in space and increase in speed suggests that despite the advanced technological capabilities of CAVs, complete automation might lead to more aggressive driving behaviors that could make vulnerable road users feel less safe. The narrowed interaction space of 0.3 m at 100 % CAV penetration is particularly concerning as it provides minimal margin for error and could create anxiety for pedestrians using the crosswalk. This observation aligns with the safety deterioration phase shown in the graph, where crash numbers slightly increase after the optimal 70 % penetration rate, potentially due to this more aggressive spatial behavior of CAVs when they dominate the traffic environment.

On the other hand, the remaining unavoidable collisions were primarily due to intentional rule-breakers, which is extremely hard for CAVs to react especially when the penetration rate is above 80 %. Typical examples are provided in Fig. 8e. The first scenario is a motorcyclist deliberately running a red light, creating a hazardous situation as CAV fleets approach at 33 km/h. The motorcyclist's behavior represents a direct violation of traffic rules, forcing the CAV system to respond to an unexpected and dangerous situation at a signalized intersection. The second scenario presents a particularly dangerous situation where two children suddenly emerge from behind parked vehicles into driving lanes where CAV fleets are traveling at 30 km/h. This creates a significant visibility and prediction challenge for the ADS. Such unexpected movements into the driving lane constitute a form of rule-breaking behavior that might induce the limitation of ADS's safety protocols. From the perspective of CAV integration, such crash data suggests that replacing all conventional vehicles with CAVs will not entirely eliminate on-road collisions in existing traffic systems.

Even under optimal conditions, traffic conflicts persist in high- and

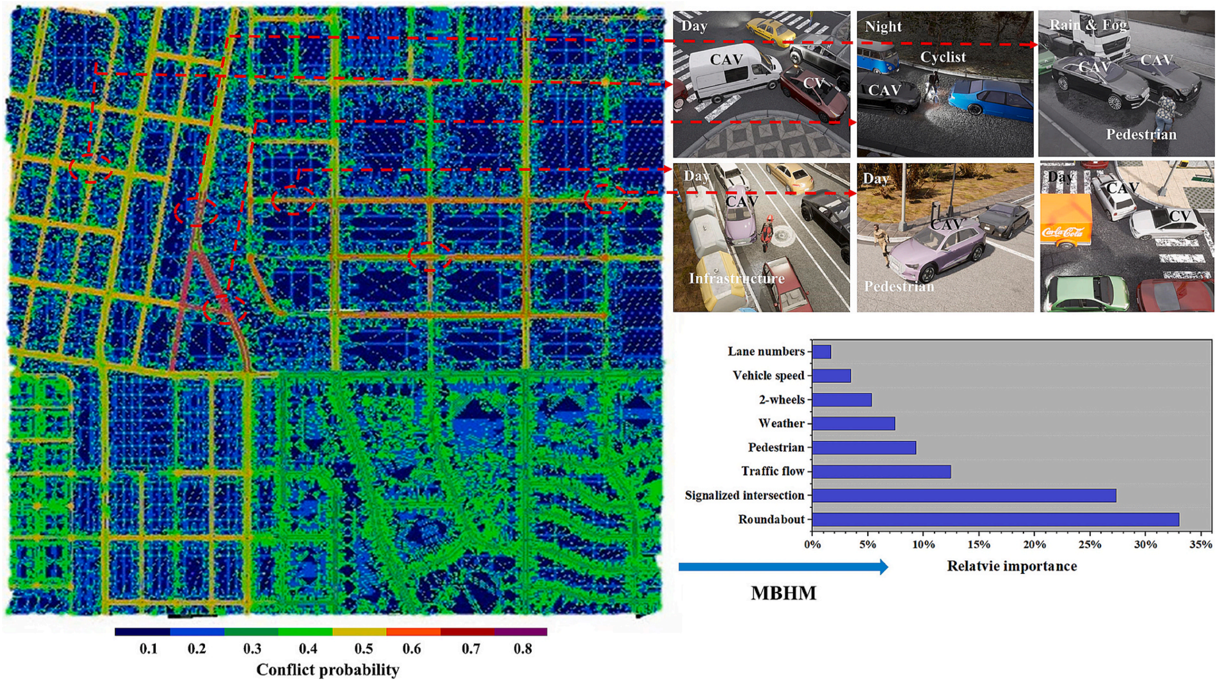

Fig. 10. The outcomes of traffic conflicts analysis under a 70% CAV penetration condition, including typical illustrative examples of observed crashes from high-risk areas and results from the MBHM.

medium-density residential areas. Fig. 10 illustrates the results of a traffic conflicts analysis under the 70 % CAV penetration scenario. A general overview from the grid-based map of the testing environment identifies the transition zone—comprising three large roundabouts connecting areas A1, A2, and A3—as the most significant conflict area, with a conflict probability greater than 0.7. The second significant conflict zones are located in the congested CBD area (A2), where CAVs are more likely to encounter conflicts at intersections and roundabouts, with a conflict probability exceeding 0.6. Overall, nearly 70 % of CAV-involved traffic conflicts occurred at roundabouts and signalized intersections. Of the total of six crashes, four occurred at intersections within areas A1 and A2. As illustrated in Fig. 10, the primary cause of these crashes was the sudden maneuver changes by vulnerable road users (noted in 4 cases). This was followed by issues arising from the incorporation between CAVs and human-driven vehicles (CVs). Most incidents (4 out of 6) occurred during fine weather conditions in daytime. In addition to deliberate rule-breaking (as shown in Fig. 8e), conflicts were primarily driven by vehicular interactions, particularly with large vehicles such as trucks at the entry and exit points of roundabouts. This was particularly evident at the smaller roundabouts located in area A3, where the conflict probability exceeded 0.5. Traffic flow conditions also present challenges to CAV safety, particularly in urban areas where lower speed limits and limited road clearance capabilities adversely affect performance. In contrast, rural areas demonstrate superior conflict clearance capabilities due to higher speed limits and lower traffic volumes, maintaining a conflict probability of approximately 0.4. Fig. 10 also includes a plot of the MBHM output, illustrating the relative importance of contributing factors to these traffic conflicts. Intersection design emerges as the most significant factor, with roundabouts and signalized intersections ranking first and second in relative importance, accounting for approximately 34 % and 27 % respectively. Conversely, environmental factors—including both static and dynamic elements—have a less substantial impact on CAV safety performance across the 20×20 km^2 testing environment.

5. Discussion

In this paper, we introduce a co-simulation-based safety evaluation framework for CAV operations within a large-scale testing environment. This framework integrates with enhanced traffic microsimulation to create a highly customizable, high-fidelity environment that closely mimics naturalistic driving scenarios. Based on such a framework, our case study is pioneering in the safety validation of CAV operations on large-scale testing tracks. The strategic use of the CARLA platform has enabled a more comprehensive assessment of CAV operations, providing deeper insights into the strategies for implementing CAVs in the near future.

One key finding from the utilization of this research tool concerns the relationship between CAV penetration rates and safety performance across a large-scale interconnected road network. Although conventional wisdom suggests that 100 % CAV penetration would lead to optimal safety outcomes (Xiao et al., 2021), our findings challenge this assumption. Previous work indicated that network performance metrics generally improve with higher CAV penetration, thanks to synchronized and robust driving behaviors (Sinha et al., 2020). However, our analysis reveals that safety benefits do not consistently follow a linear correlation with increasing penetration rates. In simpler traffic environments such as highways and rural roads (designated as area A4 in our case study), 100 % CAV penetration does indeed provide the best safety outcomes, aligning with Papadoulis et al. (2019) who reported a 90–94 % reduction in conflicts at full penetration on motorways. Conversely, in complex urban settings like CBDs and dense residential zones (areas A1 and A2 as defined in our case study), the optimal safety was observed at approximately 70 % CAV penetration. This finding is contrary to earlier assumptions of maximal safety at full automation (Papadoulis et al., 2019; Rahman et al., 2019; Sinha et al., 2020). More evidently, when assessing the operational efficiency of CAVs in a large-scale network-level traffic environment, our findings show that crash rates significantly decreased upon reaching a 60 % CAV penetration rate. Furthermore, maintaining a 70 % CAV penetration rate achieved an overall stable balance between automated and conventional human-driven vehicles. However, crash rates began to increase again at an 80 % penetration rate and continued to escalate as penetration approached 90 %, never dropped to zero even with full CAV implementation (Fig. 8 & Fig. 9). This evidence contradicts earlier expectations of linear safety improvements with increased automation levels up to 90 % penetration (Rahman et al., 2019). In our case study, the increase in crashes observed after a 70 % CAV penetration rate, with numbers rising from 6 to 7–10 crashes, can be attributed to several factors:

System homogeneity effects.

- As the system becomes more homogeneous with higher penetration rates, it may become more susceptible to synchronized behaviors. When most vehicles operate under similar algorithms, they might respond to disturbances in similar ways, potentially amplifying rather than dampening perturbations.
- The reduced diversity in driving behaviors could make the system less resilient to interactive situations. For instance, the calculated safe and efficient operating strategies of CAVs might result in less space being given to pedestrians and cyclists. This can increase the probability of vehicle–pedestrian conflicts and crashes (Fig. 9).

Complex interaction dynamics.

- At very high penetration rates, the dense network of connected vehicles might create more complex interaction patterns, particularly in response to unexpected situations. For instance, the reduced car-following distance enforced by the system's algorithms may result in insufficient space for vehicles to react appropriately to rule-breakers' behavior, even if these actions were anticipated by the ADS (Fig. 8).
- The high number of vehicles responding to the same information simultaneously could lead to unintended convergence in behavior.

These findings suggest that in complex urban networks, a high proportion of CAVs might actually escalate conflicts. They underscore the importance of maintaining a mix of human drivers in urban areas to stabilize traffic flow and enhance safety during the transition to full automation. These insights have significant implications for crafting CAV deployment strategies, highlighting the necessity of a balanced approach during the adoption phase.

Furthermore, the analysis of CAV safety performance reveals a more nuanced relationship between intersection safety in traditional and CAV-integrated traffic environments than initially apparent. According to the empirical data (Fig. 4), safety at intersections remains a critical issue for human drivers. In our case study, similar patterns in conflict and crash typology and frequency during CAV operations were also observed. This suggests that fundamental traffic safety challenges, particularly at intersections, persist regardless of the type of vehicle operation. However, despite the similarity in conflict patterns between human-driving scenarios and mixed traffic scenarios, the underlying causes and potential solutions differ. At signalized intersections, our analysis revealed that despite the presence of automated control systems, these locations continue to be significant conflict points. They rank second in terms of the relative importance of contributing factors to traffic conflicts. The inadequate incorporation between human drivers and CAVs at intersections contributes significantly to conflicts and crashes (Fig. 10). At roundabouts, an even higher conflict probability of 70 % was identified. These conflicts primarily resulted from interactions between CAV fleets and other road users, chiefly due to the significantly reduced distance between the CAV fleet and vulnerable road users (Fig. 9). The crashes at these locations were mainly attributed to intentional rule-breaking behavior (Fig. 10). There are evident instances where existing infrastructure design creates challenges to CAV operation, particularly in complex scenarios involving multiple road users. Unlike human-to-human interactions, which rely on informal communication and visual cues, CAVs interpret and respond to other road users' behavior changes based on sensor inputs and programmed actions. Nevertheless, even if rule-breaking behavior can be anticipated by the ADS within the ODD, the constrained space availability for maneuvering can limit effective accident avoidance.

These findings highlight that while intersection safety remains a pivotal concern, the approach to road safety management may need substantial revisions to better accommodate CAVs. This observation is in line with the conclusions of Virdi et al. (2019), which suggest that higher penetration rates of CAVs necessitate special considerations of infrastructure to ensure their effective integration. Concerning infrastructure design, there appears to be a fundamental mismatch between the current design philosophy and the operational needs of CAVs. For instance, road safety management strategies might need to incorporate dedicated CAV lanes to provide additional space for automated fleets, allowing CAVs to safely avoid collisions with infrastructure or other road users, particularly when encountering intentional rule-breaking behaviors. Moreover, traditional roundabouts, which present complex navigational challenges for CAV fleets, may need to be re-evaluated or re-designed to ensure effective and safe operation. In addition, regarding the inconsistency in human behavior which presents a significant challenge for CAV integration, while human drivers can adapt intuitively to unexpected behaviors, CAVs require more structured and predictable environments. This suggests that future road safety management might need to focus on creating clearer delineation of movement patterns and establishing more standardized behavioral expectations at intersections.

We acknowledge that this study has objective limitations in its methodology and scope. Firstly, the lack of systematic validation of traffic dynamics raises questions about the direct applicability of the findings. However, the small-scale validation conducted in a representative area has proven the accuracy of our methods in reproducing traffic dynamics. Given the impracticality of replicating such a large-scale testing environment in the real world—due to both safety concerns and current technological limitations—the study still provides a valuable alternative for data generation, benefiting both researchers and practitioners. One additional limitation is the simplified modelling of deliberate rule-breaking behaviors, which may not fully capture the complexity and unpredictability of real human violations. Despite this, our modelling strategies successfully incorporate randomness in road user behavior, greatly enhancing the realism of the traffic micro-simulation. Lastly, while the research covers a range of road types and conditions, it does not fully address temporary road situations such as construction zones, emergency scenarios, or special events, which may significantly impact CAV performance. These areas present opportunities for future research to further refine the models and extend their applicability.

CRediT authorship contribution statement

Zheng Xu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xiaomeng Wang:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation. **Xuesong Wang:** Writing – review & editing, Supervision, Investigation. **Nan Zheng:** Writing – review & editing, Validation, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

Abdel-Aty, M., Cai, Q., Wu, Y., Zheng, O., 2022. Evaluation of automated emergency braking system's avoidance of pedestrian crashes at intersections under occluded conditions within a virtual simulator. *Accid. Anal. Prev.* 176, 106797.

Al-Msari, H., Koting, S., Ahmed, A.N., El-Shafie, A., 2024. Review of driving-behaviour simulation: VISSIM and artificial intelligence approach. *Heliyon* 10 (4).

Alozi, A.R., Hussein, M., 2022. Evaluating the safety of autonomous vehicle–pedestrian interactions: an extreme value theory approach. *Anal. Methods Accid. Res* 35, 100230.

Astarita, V., Giofré, V.P., 2019. From traffic conflict simulation to traffic crash simulation: Introducing traffic safety indicators based on the explicit simulation of potential driver errors. *Simul. Model. Pract. Theory* 94, 215–236.

Bauer, T.P., Edinger, J. and Becker, C., 2019, March. A qualitative and quantitative analysis of real time traffic information providers. In 2019 IEEE International Conference on Pervasive Computing and Communications Workshops (PerCom Workshops) (pp. 113-118). IEEE.

Beirlant, J., Goegebeur, Y., Segers, J., Teugels, J.L., 2006. Statistics of extremes: theory and applications. John Wiley & Sons.

CARLA Simulator, 2023. Catalogue [online]. Available at: https://carla.readthedocs.io/en/0.9.15/catalogue/ [Accessed 25 September 2024].

Caird, J.K., Simmons, S.M., Wiley, K., Johnston, K.A., Horrey, W.J., 2018. Does talking on a cell phone, with a passenger, or dialing affect driving performance? An updated systematic review and meta-analysis of experimental studies. *Hum. Factors* 60 (1), 101–133.

Chen, P., Ni, H., Wang, L., Yu, G., Sun, J., 2024. Safety performance evaluation of freeway merging areas under autonomous vehicles environment using a co-simulation platform. *Accid. Anal. Prev.* 199, 107530.

Colledanchise, M., Natale, L., 2021. On the implementation of behavior trees in robotics. *IEEE Rob. Autom. Lett.* 6 (3), 5929–5936.

Corbett, C., Simon, F., 1992. Decisions to break or adhere to the rules of the road, viewed from the rational choice perspective. Brit. J. Criminol. 32, 537.

Devalla, J., 2018. Who violates traffic rules? J. Austral. College Road Safety 29 (2), 50–53.

Dosovitskiy, A., Ros, G., Codevilla, F., Lopez, A. and Koltun, V., 2017, October. CARLA: An open urban driving simulator. In Conference on robot learning (pp. 1-16). PMLR.

Federal Highway Administration, 2024. Estimated Vehicle-Miles of Travel by Region - July 2024. Available at: https://www.fhwa.dot.gov/policyinformation/travel_monitoring/24jultvt/24jultvt.pdf [Accessed 25 September 2024].

Feng, S., Feng, Y., Yan, X., Shen, S., Xu, S., Liu, H.X., 2020. Safety assessment of highly automated driving systems in test tracks: A new framework. *Accid. Anal. Prev.* 144, 105664.

Feng, S., Yan, X., Sun, H., Feng, Y., Liu, H.X., 2021. Intelligent driving intelligence test for autonomous vehicles with naturalistic and adversarial environment. *Nat. Commun.* 12 (1), 748.

Freeman, J., Rakotonirainy, A., 2015. Mistakes or deliberate violations? A study into the origins of rule breaking at pedestrian train crossings. *Accid. Anal. Prev.* 77, 45–50.

Fu, C., Sayed, T., Zheng, L., 2021. Multi-type Bayesian hierarchical modeling of traffic conflict extremes for crash estimation. *Accid. Anal. Prev.* 160, 106309.

Hansson, L., 2020. Regulatory governance in emerging technologies: The case of autonomous vehicles in Sweden and Norway. *Res. Transp. Econ.* 83, 100967.

Jin, I.G., Avedisov, S.S., He, C.R., Qin, W.B., Sadeghpour, M., Orosz, G., 2018. Experimental validation of connected automated vehicle design among human-driven vehicles. *Transp. Res. Part C Emerging Technol.* 91, 335–352.

Khan, F., Falco, M., Anwar, H., Pfahl, D., 2023. Safety testing of automated driving systems: a literature review. *IEEE Access*.

Kolekar, S., Petermeijer, B., Boer, E., de Winter, J., Abbink, D., 2021. A risk field-based metric correlates with driver's perceived risk in manual and automated driving: A test-track study. *Transp. Res. Part C Emerging Technol.* 133, 103428.

Koopman, P., Wagner, M., 2016. Challenges in autonomous vehicle testing and validation. SAE *Int. J. Transport. Safety* 4 (1), 15–24.

Lee, D., Hess, D.J., 2020. Regulations for on-road testing of connected and automated vehicles: Assessing the potential for global safety harmonization. *Transp. Res. A Policy Pract.* 136, 85–98.

Li, G., Yang, Y., Li, S., Qu, X., Lyu, N., Li, S.E., 2022. Decision making of autonomous vehicles in lane change scenarios: Deep reinforcement learning approaches with risk awareness. *Transp. Res. Part C Emerging Technol.* 134, 103452.

Liu, Q., Wang, X., Liu, S., Yu, C., Glaser, Y., 2024. Analysis of pre-crash scenarios and contributing factors for autonomous vehicle crashes at intersections. *Accid. Anal. Prev.* 195, 107383.

Malik, F.M., Khattak, H.A., Almogren, A., Bouachir, O., Din, I.U., Altameem, A., 2020. Performance evaluation of data dissemination protocols for connected autonomous vehicles. *IEEE Access* 8, 126896–126906.

Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A.A., Veness, J., Bellemare, M.G., Graves, A., Riedmiller, M., Fidjeland, A.K., Ostrovski, G., Petersen, S., 2015. Human-level control through deep reinforcement learning. *Nature* 518 (7540), 529–533.

Ni, H., Yu, G., Chen, P., Zhou, B., Liao, Y., Li, H., 2024. An integrated framework of lateral and longitudinal behavior decision-making for autonomous driving using reinforcement learning. IEEE Transactions on Vehicular Technology.

Papadoulis, A., Quddus, M., Imprialou, M., 2019. Evaluating the safety impact of connected and autonomous vehicles on motorways. *Accid. Anal. Prev.* 124, 12–22.

Pereira, J., Premebida, C., Asvadi, A., Cannata, F., Garrote, L. and Nunes, U.J., 2019, June. Test and evaluation of connected and autonomous vehicles in real-world scenarios. In 2019 IEEE Intelligent Vehicles Symposium (IV) (pp. 14-19). IEEE.

Rahman, M.M., Thill, J.C., 2023. Impacts of connected and autonomous vehicles on urban transportation and environment: A comprehensive review. *Sustain. Cities Soc.* 96, 104649.

Rahman, M.S., Abdel-Aty, M., Lee, J., Rahman, M.H., 2019. Safety benefits of arterials' crash risk under connected and automated vehicles. *Transp. Res. Part C Emerging Technol.* 100, 354–371.

Rana, M.M., Hossain, K., 2023. Connected and autonomous vehicles and infrastructures: A literature review. *Int. J. Pavement Res. Technol.* 16 (2), 264–284.

Ranjane, H., Talware, R., Bang, A., Vartak, R. and Patil, M., 2023, October. Automated System for Traffic Control and Management. In 2023 7th International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud)(I-SMAC) (pp. 1177-1184). IEEE.

2018 SAE International., 2018. Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles (J3016_201806). [online] Available at: https://www.sae.org/standards/content/j3016_201806/ [Accessed 23 October 2024].

Siebke, C., Mai, M., Prokop, G., 2022. What do traffic simulations have to provide for virtual road safety assessment? Human error modeling in traffic simulations. *IEEE Trans. Intell. Transp. Syst.* 24 (2), 1419–1436.

Simmons, S.M., Hicks, A., Caird, J.K., 2016. Safety-critical event risk associated with cell phone tasks as measured in naturalistic driving studies: A systematic review and meta-analysis. *Accid. Anal. Prev.* 87, 161–169.

Sinha, A., Chand, S., Wijayaratna, K.P., Virdi, N., Dixit, V., 2020. Comprehensive safety assessment in mixed fleets with connected and automated vehicles: A crash severity and rate evaluation of conventional vehicles. *Accid. Anal. Prev.* 142, 105567.

Sohrabi, S., Khodadadi, A., Mousavi, S.M., Dadashova, B., Lord, D., 2021. Quantifying the automated vehicle safety performance: A scoping review of the literature, evaluation of methods, and directions for future research. *Accid. Anal. Prev.* 152, 106003.

Sørensen, J.V., Ma, Z., Jørgensen, B.N., 2022. Potentials of game engines for wind power digital twin development: an investigation of the Unreal Engine. *Energy Informatics* 5 (Suppl 4), 39.

Sun, C., Leng, J., Lu, B., 2022. Interactive left-turning of autonomous vehicles at uncontrolled intersections. *IEEE Trans. Autom. Sci. Eng.* 21 (1), 204–214.

Tang, S., Zhang, Z., Zhang, Y., Zhou, J., Guo, Y., Liu, S., Guo, S., Li, Y.F., Ma, L., Xue, Y., Liu, Y., 2023. A survey on automated driving system testing: landscapes and trends. *ACM Trans. Softw. Eng. Methodol.* 32 (5), 1–62.

Tomtom Developers (2022) Documentation for Tom Tom Traffic API. [Online] Available at: https://developer.tomtom.com/traffic-api/documentation/productinformation/introduction (Accessed: 1 February 2025).

Virdi, N., Grzybowska, H., Waller, S.T., Dixit, V., 2019. A safety assessment of mixed fleets with connected and autonomous vehicles using the surrogate safety assessment module. *Accid. Anal. Prev.* 131, 95–111.

Wiedemann, R., Reiter, U., 1992. Microscopic traffic simulation: the simulation system MISSION, background and actual state. Project ICARUS (V1052) Final Report 2, 1–53.

Wijayaratna, K.P., Cunningham, M.L., Regan, M.A., Jian, S., Chand, S., Dixit, V.V., 2019. Mobile phone conversation distraction: understanding differences in impact between simulator and naturalistic driving studies. *Accid. Anal. Prev.* 129, 108–118.

Wirth, R., Foerster, A., Herbort, O., Kunde, W., Pfister, R., 2018. This is how to be a rule breaker. *Adv. Cogn. Psychol.* 14 (1), 21.

Xiao, G., Lee, J., Jiang, Q., Huang, H., Abdel-Aty, M., Wang, L., 2021. Safety improvements by intelligent connected vehicle technologies: a meta-analysis considering market penetration rates. *Accid. Anal. Prev.* 159, 106234.

Xu, C., Ding, Z., Wang, C., Li, Z., 2019. Statistical analysis of the patterns and characteristics of connected and autonomous vehicle involved crashes. *J. Saf. Res.* 71, 41–47.

Xu, Z., Jiang, Z., Wang, G., Wang, R., Li, T., Liu, J., Zhang, Y., Liu, P., 2021. When the automated driving system fails: dynamics of public responses to automated vehicles. *Transp. Res. Part C Emerging Technol.* 129, 103271.

Xu, Z., Fang, Y., Zheng, N., Vu, H.L., 2022. Analyzing the inconsistency in driving patterns between manual and autonomous modes under complex driving scenarios with a VR-enabled simulation platform. *J. Intelligent Connected Vehicles* 5 (3), 215–234.

Xu, Z., Wang, C., Jiang, T., Liu, Z., Zheng, N., 2024. Impediments to environmental awareness in autonomous driving systems and its effect on user adoption. IEEE Transactions on Intelligent Vehicles.

Xu, Z., Zheng, N., Logan, D.B., Vu, H.L., 2023. Assessing bicycle-vehicle conflicts at urban intersections utilizing a VR integrated simulation approach. *Accid. Anal. Prev.* 191, 107194.

Xu, Z., Zheng, N., Lv, Y., Fang, Y., Vu, H.L., 2024a. Analyzing scenario criticality and rider's intervention behavior during high-level autonomous driving: A VR-enabled approach and empirical insights. *Transp. Res. Part C Emerging Technol.* 158, 104451.

Xu, Z., Zheng, Z., Xiao, D., Tu, R., Ma, W., Zheng, N., 2024b. Assessing the impact of passenger compliance behavior in CAVs on environmental benefits. *Transp. Res. Part D: Transp. Environ.* 133, 104278.

Yan, X., Zou, Z., Feng, S., Zhu, H., Sun, H., Liu, H.X., 2023. Learning naturalistic driving environment with statistical realism. *Nat. Commun.* 14 (1), 2037.

Yang, S., Huang, Y., Li, L., Feng, S., Na, X., Chen, H., Khajepour, A., 2024. How to guarantee driving safety for autonomous vehicles in a real-world environment: a perspective on self-evolution mechanisms. *IEEE Intell. Transp. Syst. Mag.*

Yu, R., Li, S., 2022. Exploring the associations between driving volatility and autonomous vehicle hazardous scenarios: Insights from field operational test data. *Accid. Anal. Prev.* 166, 106537.